testbook PASS PRO MAX

General Knowledge
World Geography

STUDY NOTES

Mapping: Waterfalls

Mapping: Waterfalls

Introduction to Waterfalls

A Waterfalls are some of nature's most spectacular features, formed when water flows over a vertical drop or a cliff. There are several types of waterfalls, each with distinct characteristics based on the terrain, water flow, and erosion patterns. Here's an overview of waterfalls, their types, and key characteristics:

Types of Waterfalls

1. **Cataract**
   - **Description**: A large, powerful waterfall, often with a very wide flow of water. Cataracts are usually associated with high-volume waterfalls.
   - **Example**: **Victoria Falls** (on the Zambezi River, between Zambia and Zimbabwe).
2. **Plunge Waterfall**
   - **Description**: A waterfall where the water drops vertically from a great height, typically without touching the cliff face during its fall.
   - **Example**: **Angel Falls** in Venezuela (the highest uninterrupted waterfall in the world).
3. **Cascade Waterfall**
   - **Description**: A series of small waterfalls or steps, where water flows over multiple ledges or rocks, creating a cascading effect.
   - **Example**: **Multnomah Falls** in Oregon, USA.
4. **Block Waterfall**
   - **Description**: A waterfall where the water flows over a wide, flat surface, and drops uniformly across the entire width of the river or stream.
   - **Example**: **Niagara Falls** in North America.
5. **Tiered Waterfall**
   - **Description**: A waterfall composed of multiple drops or stages, where water cascades from one level to another.
   - **Example**: **Jog Falls** in India, one of the highest plunge waterfalls in India.
6. **Segmented Waterfall**
   - **Description**: A waterfall that is divided into several distinct segments or streams of water, often because of varying rock formations or branching streams.
   - **Example**: **Bridalveil Falls** in Yosemite National Park, USA.
7. **Fan Waterfall**
   - **Description**: A waterfall where the water spreads out in a fan-like shape as it falls, often due to the shape of the cliff or the flow of water.
   - **Example**: **Ramnefjellsfossen** in Norway.
8. **Horsetail Waterfall**
   - **Description**: A waterfall where the flow of water maintains contact with the rock face, resembling a horse's tail.
   - **Example**: **Yosemite Falls** in California, USA.

Key Characteristics of Waterfalls

1. **Height**
   - The height of a waterfall is one of the defining features, often measured from the top of the fall to the base. Some waterfalls, like **Angel Falls**, have extreme vertical drops, making them some of the tallest on Earth.
2. **Flow Rate**
   - This refers to the volume of water flowing over the waterfall. High flow rates are seen in larger rivers, while smaller streams may have a more gentle flow.
3. **Width**
   - The width of a waterfall can vary dramatically. For example, **Niagara Falls** is known for its broad width, while others, like **Sutherland Falls** in New Zealand, are narrower.
4. **Rock Formation**
   - The type of rock and its erosion characteristics play a crucial role in the waterfall's formation. Softer rocks erode faster than harder rocks, which can create stepped or tiered formations.
5. **Water Source**
   - Waterfalls can form from rivers, streams, glaciers, or underground springs, each source contributing to different types of waterfalls.
6. **Erosion Patterns**
   - The process of erosion by the flowing water over time shapes the waterfall. The force of water can carve out deeper pools at the base of waterfalls, leading to features like plunge pools or gorges.
7. **Seasonal Variations**
   - Waterfalls may experience changes in flow depending on the season. During the rainy season, waterfalls may swell with increased water flow, while in the dry season, the flow may reduce or even stop.
8. **Plunge Pools**
   - Many waterfalls create deep pools at their base due to the force of the water hitting the ground, a phenomenon known as a **plunge pool**. These pools are often characterized by swirling currents.
9. **Mist and Rainbows**
   - Large waterfalls often generate mist, which can lead to the formation of rainbows when sunlight interacts with the mist droplets.

How Waterfalls Form

Waterfalls are formed through processes like erosion, weathering, and geological activity. Here's a basic breakdown:

1. **Erosion**:
   Over time, water flowing over rocks erodes softer materials, creating a drop where the harder rock remains. The water continues to erode the drop, causing it to deepen and widen.

2. **Retreat**:
   As the waterfall erodes the rock at its base, it gradually retreats upstream, leaving behind a gorge or canyon.
3. **Differential Erosion**:
   Waterfalls often form in places where the river encounters layers of rock with different hardness. Softer rocks erode faster, causing the harder rocks to form a ledge or drop.

Waterfall	Locality	Country	River
Angel Falls	Canaima National Park, Bolívar	Venezuela	Carrao River
Tugela Falls	KwaZulu-Natal	South Africa	Tugela River
Tres Hermanas Falls	Otishi National Park, Junin	Peru	Pangoa River
Olo'upena Falls	Molokai, Hawaii	United States	Unnamed Stream
Yumbilla Falls	Amazonas	Peru	Yumbilla River
Vinnufossen	Møre og Romsdal	Norway	Vinnu River
Balåifossen	Hordaland	Norway	Bala River
Pu'uka'oku Falls	Hawaii	United States	Unnamed Stream
James Bruce Falls	British Columbia	Canada	Unnamed Stream
Browne Falls	Doubtful Sound, Southland	New Zealand	Deep Cove
Strupenfossen	Sogn og Fjordane	Norway	Strupen River
Ramnefjellsfossen	Sogn og Fjordane	Norway	Ramnefjell River
Waihilau Falls	Hawaii	United States	Waihilau Stream
Colonial Creek Falls	Washington	United States	Colonial Creek
Mongefossen	Møre og Romsdal	Norway	Monge River
Gocta Cataracts	Amazonas	Peru	Gocta River
Mutarazi Falls	Manicaland	Zimbabwe	Mutarazi River
Kjelfossen	Sogn og Fjordane	Norway	Kjelfoss River
Yosemite Falls	California	United States	Yosemite Creek
Cascades de Trou de Fer	Réunion	France	Rivière de l'Est
Olmaafossen	Møre og Romsdal	Norway	Olmaafoss River

testbook PASS PRO MAX

Manawainui Falls	Hawaii	United States	Manawainui Stream
Kjeragfossen	Rogaland	Norway	Kjerag River
Avalanche Basin Falls	Montana	United States	Avalanche Creek
Harrison Basin Falls	Montana	United States	Harrison Creek
Haloku Falls	Hawaii	United States	Haloku Stream
Chamberlain Falls	Southland	New Zealand	Chamberlain Creek
Alfred Creek Falls	British Columbia	Canada	Alfred Creek
Døntefossen	Møre og Romsdal	Norway	Dønte River
Brufossen	Hordaland	Norway	Brufoss River
Spirefossen	Sogn og Fjordane	Norway	Spire River
Lake Unknown Falls	Otago	New Zealand	Unnamed Stream
Kukenam, Salto	Bolívar	Venezuela	Kukenam River
Yutaj, Salto	Amazonas	Venezuela	Yutaj River
La Chorrera Falls	Cundinamarca	Colombia	Quebrada Chorrera

World's Highest Artificial Waterfalls

Artificial Waterfall	Height (Metres/Feet)	Location	Country
Cascata Delle Marmore (Created by Ancient Romans)	165 metres (541 ft)	Parco Fluviale del Nera	Italy
Liebian Waterfall (Tallest waterfall in a building)	108 metres (354 ft)	Liebian International Plaza	China
Rain Vortex (Largest Indoor Waterfall)	40 metres (131 ft)	Jewel Changi Airport	Singapore
Cloud Forest Indoor Waterfall	35 metres (115 ft)	Cloud Forest, Gardens by the Bay	Singapore
Jurong Falls	30 metres (98 ft)	Jurong Bird Park	Singapore

Viktoriapark Waterfall	24 metres (79 ft)	Viktoriapark	Germany
Lotte Music Fountain	18.2 metres (60 ft)	Aqua Mall, Lotte Department Store (Busan)	South Korea